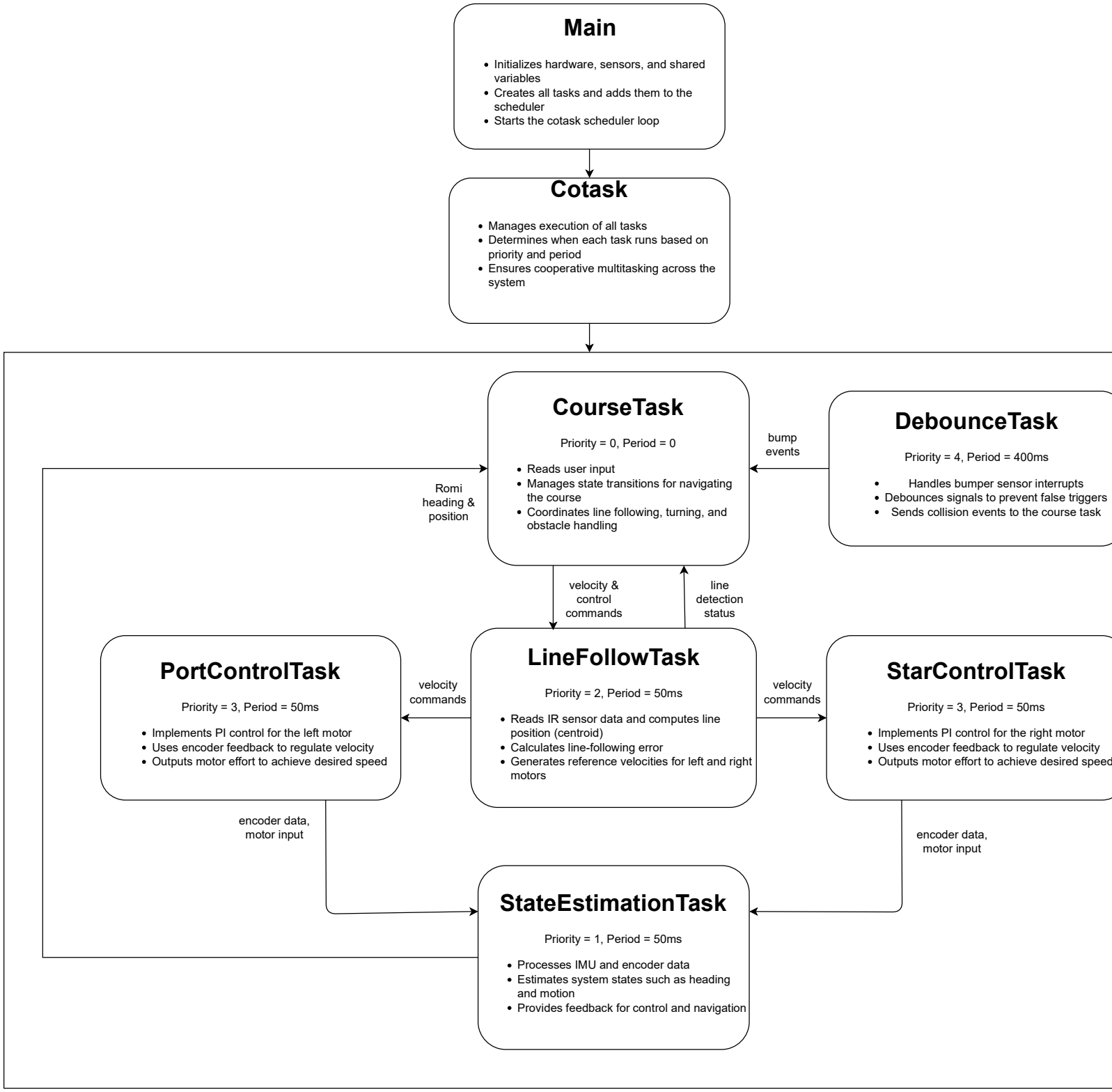
Main
Initializes hardware, sensors, and shared variables
Creates all tasks and adds them to the scheduler
Starts the cotask scheduler loop
Cotask
Manages execution of all tasks
Determines when each task runs based on priority and period
Ensures cooperative multitasking across the system
CourseTask
Priority = 0, Period = 0
Reads user input
Manages state transitions for navigating the course
Coordinates line following, turning, and obstacle handling
DebounceTask
Priority = 4, Period = 400ms
Handles bumper sensor interrupts
Debounces signals to prevent false triggers
Sends collision events to the course task
bump events
Romi heading & position
velocity & control commands
line detection status
PortControlTask
Priority = 3, Period = 50ms
Implements PI control for the left motor
Uses encoder feedback to regulate velocity
Outputs motor effort to achieve desired speed
velocity commands
LineFollowTask
Priority = 2, Period = 50ms
Reads IR sensor data and computes line position (centroid)
Calculates line-following error
Generates reference velocities for left and right motors
velocity commands
StarControlTask
Priority = 3, Period = 50ms
Implements PI control for the right motor
Uses encoder feedback to regulate velocity
Outputs motor effort to achieve desired speed
encoder data, motor input
encoder data, motor input
StateEstimationTask
Priority = 1, Period = 50ms
Processes IMU and encoder data
Estimates system states such as heading and motion
Provides feedback for control and navigation